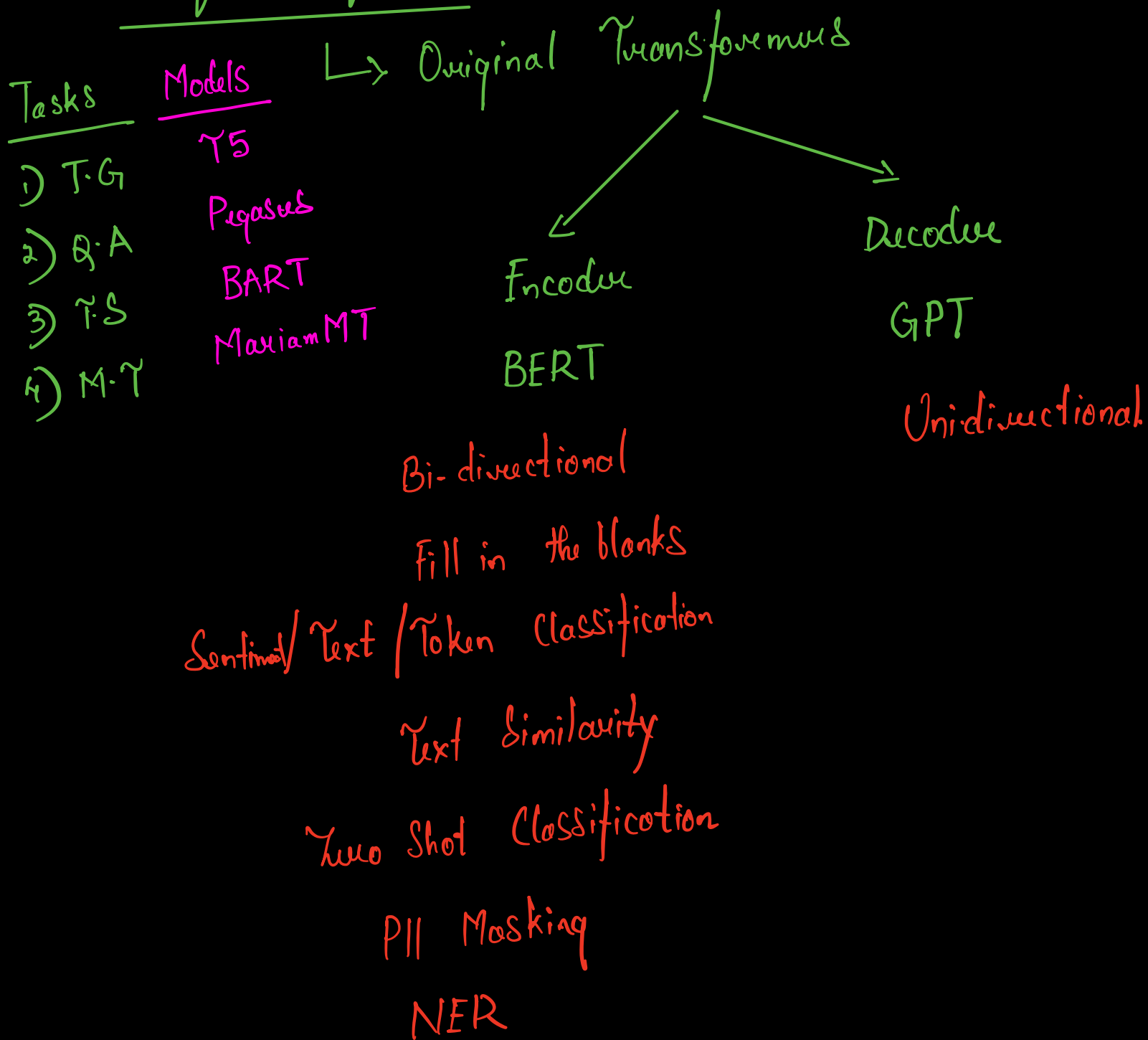


Seq 2 Seq



(BERT)

1) Bidirectional Training
L | R Context Simultaneously

MHA

Masked MHA

* No Generation *

1) Deeper Understanding

2) Context Enrichment

3) Task Versatility

4) Transfer learning (Fine Tune on Pre Trained)

Tokenization Scheme

BERT → Word Piece

GPT → Byte Pair Encoding

LLAMA/T5 → Sentence Piece

(Word Piece) → most frequent pairs

merge
likelihood

Byte Pair Encoding

Unbelievable

Un + believe + able

We must break text into smaller pieces.

Storing Full words $\rightarrow$ HUGE Vocab

Store Every character $\rightarrow$ Sentences becomes very long

BPE $\rightarrow$ Byte Pair Encoding

Build words by repetition or merging pairs

[low]

lowest
newer
wider } → Tiny Dataset

Step 1 → Split by characters

l	o	w			
l	o	w	e	s	t
n	e	w	e	r	
w	i	d	e	r	

Step 2

Count the most common appearing pairs.

l o
1 + 0 ⇒ 10

l	o	w			
l	o	w	e	s	t
n	e	w	e	r	
w	i	d	e	r	

⇒ 10 + w = low

low				
low	e	s	t	
n	e	w	e	u
w	i	d	e	u

$$e + u = eu$$

low				
low	e	s	t	
n	e	w	ee	
w	i	d	uu	

Final Estimation

low
ee
est
new
wid

Powerful

✓✓ unpredictable

un + predict + able

BERT WordPiece

BPE

GPT, Roberta

Merge most frequent pairs

WordPiece

BERT

Merge pairs
based on
maximum
likelihood.

Word Piece (M.L.F.)

$$\text{Score}(\text{pair}) = \frac{\text{freq}(\text{pair})}{\text{freq}(\text{first}) \times \text{freq}(\text{seq})}$$

Simple Dataset

playing

played

player

⊕ $p + l + a + y$

$pl + a$

$pla + y$

play

Pairs:-

pl la ay
ed we

play

Rough Vocab Size ≈ 30000

[UNK] $\Rightarrow$ unknown words

unhappiness

BPE :- un + happy + ness

Word :- un + ##happy + ##ness

→ continuation.

[UNK]